

# Adyapan Hospital Management System (HMS)

Complete Architectural Guide, File Structure & Workflow Blueprint

## 1. Executive Summary & Tech Stack

Adyapan HMS is an enterprise-grade, multi-tenant digital healthcare platform designed for outpatient departments (OPD), consultation workstations, queue management, electronic prescriptions (Rx), pharmacy dispensing, and patient billing.

- **Backend Architecture:** Node.js (v24 / ESM), Express.js REST API, Socket.IO real-time engine, PDFKit document generation.
- **Database Layer:** PostgreSQL 18, Prisma ORM (v6.19), Embedded PostgreSQL runner for zero-dependency local execution.
- **Frontend Architecture:** React 18 SPA, Vite 5 build tool, Tailwind CSS, Lucide React icons, Axios HTTP interceptors.
- **Multi-Tenancy Isolation:** Every clinical, administrative, and financial entity is strictly scoped by hospitalId.
- **Real-Time WebSockets:** Instant queue status progression, live doctor workstation calling, and TV waiting-room board broadcasts.

## 2. Complete Directory Structure

```
HMS/
% % % backend/
%   % % % .env
%   % % % prisma/
%   %   % % schema.prisma
%   %   % % seed.js
%   % % % src/
%       % % % app.js
%       % % % server.js
%       % % % config/
%       % % % middlewares/
%       % % % modules/
%       % % % scripts/
%       % % % utils/
%
% % % frontend/
% % % vite.config.js
% % % src/
% % % App.jsx
% % % context/
% % % routes/
% % % services/
% % % components/common/
% % % pages/

# Express REST API + Prisma ORM
# Active environment variables (Port 5000, DB: 5433)

# PostgreSQL relational schema (15+ models)
# Automated seeder (Hospitals, Doctors, Meds)

# Express app pipeline & global error handler
# HTTP + Socket.IO server entry point
# Database (db.js) & Env singletons
# JWT Auth, RBAC Authorization, Rate Limiter
# 17 Business Logic Modules (Route/Ctrl/Service)
# Embedded PostgreSQL runner (pgServer.js)
# Generators (UHID, Prescription, Invoice, PDF)

# React 18 Single Page Application
# Vite dev server (Port 5173) & /api proxy

# Client Router & Role Protected Routes
# AuthContext (JWT storage, active user state)
# ProtectedRoute guard for RBAC
# Axios API client & WebSockets socket.js
# Shell Layout, Header, and Role Dynamic Sidebar
# 14 Clinical, Queue, Pharmacy & Billing Desks
```

# Backend Architecture & Core Modules

Controller-Service-Route Design Pattern

## 3. Server Core & Middlewares

- **src/server.js:** Binds Express app with Socket.IO HTTP server on port 5000. Handles graceful shutdown signals (SIGINT, SIGTERM).
- **src/app.js:** Configures Helmet security headers, CORS origin whitelisting, JSON parsing, Morgan logging, and central error handling.
- **src/middlewares/auth.js:** authenticate verifies Bearer JWT token and attaches req.user. authorize(...roles) enforces strict role-based access control.
- **src/middlewares/errorHandler.js:** Translates Prisma exceptions (e.g. P2002 unique constraint, P2025 record not found) into clean HTTP JSON responses.
- **src/scripts/pgServer.js:** Starts embedded PostgreSQL 18 cluster on port 5433 storing persistent data in .pgdata/.

## 4. Feature Modules Breakdown

Each feature in backend/src/modules/ follows a strict separation of concerns:

- **modules/auth:** Login authentication, bcrypt password validation, JWT issuance, profile lookup, and demo quick-role picker.
- **modules/users:** Hospital staff management, onboarding new doctors/nurses/admins, assigning doctor clinical specializations.
- **modules/departments:** Clinical department catalog (Cardiology, Neurology, Pediatrics, etc.) and examination room mapping.
- **modules/doctors:** Doctor profiles, consultation fees, live availability toggle (AVAILABLE, BUSY, ON\_LEAVE), and weekly shifts.
- **modules/patients:** Comprehensive registration with duplicate phone detection, quick walk-in registration, and longitudinal EMR.
- **modules/appointments:** Pre-booked scheduling, conflict detection, time slot calculation, rescheduling, and cancellations.
- **modules/tokens:** Issues real-time queue tokens (T-001, W-002) for walk-ins and converts scheduled appointments upon arrival.
- **modules/queue:** Queue manager: "Call Next Patient", live queue stats, TV waiting-room board broadcasts, and token reassignments.
- **modules/consultations:** Doctor workstation: clinical notes, symptoms, diagnosis, vitals capture (BP, Pulse, SpO2, BMI), and closure.
- **modules/prescriptions:** Digital Rx orders, hospital formulary lookup, on-the-fly custom drug addition, auto-qty calculations, PDF generator.
- **modules/pharmacy:** Inventory catalog, drug batches with expiry dates, FIFO dispensing against prescriptions, and POS sales receipts.
- **modules/billing:** Automatic invoice generation from consultations, recording cash/card/UPI payments, refund ledger, receipts.
- **modules/reports:** Statistical aggregations: financial revenue, department workloads, patient demographics, and CSV export.
- **modules/notifications:** In-app alert bell notifications and external communication dispatch hooks.
- **modules/realtime:** Socket.IO gateway managing hospital-specific rooms for real-time queue advancement and audio chimes.

# Database Architecture & Prisma Schema

15 Interconnected Relational Models in PostgreSQL

## 5. Core Database Models (prisma/schema.prisma)

- **Hospital:** Tenant root model. All clinical and administrative entities belong to a hospitalId.
- **User:** System login accounts. Stores email, hashed password, role (ADMIN, DOCTOR, RECEPTIONIST, etc.).
- **Department:** Hospital medical specialties (e.g. Neurology, Cardiology, General Surgery) and consultation rooms.
- **Doctor:** Physician profile linked 1-to-1 with User. Stores specialization, consultation fee, room number, status.
- **DoctorSchedule:** Weekly recurring shifts by day-of-week (0-6), start time, end time, and max slot limits.
- **Patient:** Master patient demographic and clinical record with unique UHID (ADY-YYYYMM-XXXX) and indexed phone.
- **Appointment:** Pre-booked patient visit slots with appointmentDate, timeSlot, status (BOOKED, CHECKED\_IN, CANCELLED).
- **Token:** Real-time queue ticket (e.g. T-001) tracking status (WAITING, CALLED, IN\_CONSULTATION, COMPLETED).
- **Consultation:** Doctor encounter recording symptoms, diagnosis, physical vitals (BP, Pulse, SpO2), advice, and notes.
- **Prescription:** Standardized medical prescription order (RX-YYYYMM-XXXX) linked to Patient and Consultation.
- **PrescriptionItem:** Medication item details: dosage, daily frequency (1-0-1), duration, instructions, and prescribed qty.
- **Medicine:** Hospital formulary drug catalog with unit, reorder threshold alert, and active status.
- **MedicineBatch:** Inventory batch tracking with batch number, expiration date, cost price, and stock count.
- **PharmacyDispense:** Medication fulfillment ledger recording dispensed batches, dispensing pharmacist, and total cost.
- **Invoice & Payment:** Financial ledger: tracks consultation & drug charges, tax, payment status (PAID, PARTIAL), and receipts.

## 6. Unique Identifier Generators

- **UHID (uhidGenerator.js):** Format: ADY-YYYYMM-XXXX (e.g. ADY-202609-0012). Scoped to hospital and month with collision checks.
- **Prescription Code (prescriptionCodeGenerator.js):** Format: RX-YYYYMM-XXXX. Official digital medical prescription tracking number.
- **Invoice Number (invoiceNumberGenerator.js):** Format: INV-YYYYMM-XXXX. Official billing and tax invoice identifier.

# Frontend Architecture & Patient Journey

React 18 Single Page Application & Workflow Walkthrough

## 7. Frontend Shell & Navigation

- **src/App.jsx:** Client router configuring ProtectedRoute wrappers and role authorization guards.
- **src/context/AuthContext.jsx:** Global React context storing user token, role permissions, and active doctor profile.
- **src/services/api.js:** Axios client automatically attaching JWT tokens and formatting unwrapped API response payloads.
- **src/components/common/Sidebar.jsx:** Role-tailored navigation showing only tabs permitted for Doctor, Receptionist, Pharmacist, Admin, etc.

## 8. Key Clinical & Operational Desks

- **Tokens/WalkInTokenModal.jsx:** Reception walk-in desk with quick patient creation, doctor assignment, and thermal pass printing.
- **Queue/LiveQueue.jsx & QueueTvDisplay.jsx:** Interactive queue manager and fullscreen waiting-room TV display with audio call chimes.
- **Consultations/ConsultationWorkstationModal.jsx:** Full clinical workstation displaying patient history, physical vitals, diagnoses, and digital Rx.
- **Prescriptions/CreatePrescriptionModal.jsx:** Standardized e-prescription authoring desk with formulary catalog autocomplete and auto-qty.
- **Pharmacy/DispenseModal.jsx:** Pharmacy fulfillment desk deducting drug batches FIFO and producing POS sales receipts.
- **Billing/BillingDashboard.jsx:** Invoice generator, multi-channel payment recorder (Cash/Card/UPI), and financial reports.

## 9. End-to-End Patient Journey Flow

1. RECEPTION CHECK-IN:  
Patient arrives -> WalkInTokenModal.jsx -> Issues Token #T-001 (WAITING)  
Auto-generates UHID (ADY-202609-0001) -> WebSockets updates TV display
2. QUEUE & CONSULTATION:  
Doctor clicks "Call Next Patient" -> Token status becomes CALLED -> IN\_CONSULTATION  
Doctor opens ConsultationWorkstationModal.jsx -> Records Vitals, Symptoms, Diagnosis
3. DIGITAL PRESCRIPTION (Rx):  
Doctor authors medications -> Creates official Prescription RX-202609-0001  
Auto-calculates quantity (e.g. 1-0-1 for 5 days = 10 tablets) -> Emits Pharmacy alert
4. PHARMACY DISPENSE & BILLING:  
Pharmacist fulfills prescription in DispenseModal.jsx -> Depletes inventory batches  
Cashier records payment (Cash/UPI) -> Consultation closed & Token marked COMPLETED















